



University of
Nottingham
UK | CHINA | MALAYSIA

Bluesky Brand Analytics: A Snapshot Comparison of Wired and The Verge

Student ID: 20811152

Module Code: BUSI4370 – Analytics Specializations and
Applications

Academic Year: 2025-26

Submission Date: April 20, 2026

Word Count: 2796

AI Declaration

I would like to be transparent about the involvement of generative AI tools for this coursework.

Tools Used: I used Claude AI (Anthropic) and ChatGPT for debugging the code, organizing the report, and proofreading.

How it was used: The AI tools helped me refine the code, suggest ideas for enhancing the visualization, and provide feedback on how the report can be made easier to read. The analysis, models, understanding, and business recommendations are my own.

Declaration: I am aware that this work will be treated as mine, and I accept full responsibility for what is presented in this report.

1. Executive Summary

This pilot study, commissioned by **KantarIQ**, profiles public perception of two competing technology-journalism brands on Bluesky: **Wired** and **The Verge**. A Python-driven *atproto* collection pipeline retrieved **2,731 brand-relevant posts** (1,571 Wired, 1,153 The Verge) covering September 2023–April 2026, alongside 3,124 author profiles.

Six quantitative techniques were applied: engagement profiling, VADER sentiment scoring, logistic-regression modelling of engagement drivers, Latent Dirichlet Allocation (LDA) topic modelling, keyword co-occurrence networks, and eigenvector-centrality user analysis. Outputs feed a data-derived perceptual map and a metric-justified micro-influencer recommendation per brand.

Headline findings: (i) The Verge generates more positive average sentiment (+0.137 vs +0.078) but Wired captures **2.4× the mean engagement** (23.1 vs 9.7 likes/post), driven disproportionately by negative, politically charged investigative reporting (mean 45 likes per negative Wired post vs 17 per positive). (ii) Hashtags reduce the odds of a high-engagement post by **82%** (logit OR=0.18, $p<0.001$) – a counter-intuitive but actionable result. (iii) The analysis suggests that topic spaces are mostly orthogonal: Wired leans towards power, surveillance, and Trump/Musk coverage; The Verge covers Apple, Google, Xbox, and AI gadget reviews. **Suggestions for social media users** are @thiccreese.bsky.social (Reece, 18.3K subscribers, +0.51 polarity) for Wired and @papapishu.bsky.social (Chris Person, 30.5K followers, 214 likes/post) for The Verge.

2. Methodology

The analysis was implemented in a single reproducible Jupyter notebook (*ASA_BUSI4370_20811152.ipynb*) using *atproto*, *pandas*, *nltk*, *vaderSentiment*, *gensim*, *networkx* and *statsmodels*. The pipeline progresses through five stages.

Stage 1 - Collection. A precision-first query strategy combined high-precision domain/handle terms (*wired.com*, *@theverge.com*) with recall-extending phrase queries ("*wired magazine*", "*according to the verge*"). Each query was paginated through *app.bsky.feed.searchPosts* up to a 500-post ceiling and de-duplicated by URI. This avoided the lexical-collision problem that ruled out an initial Hugging Face / Replicate pair, where *replicate* as a verb produced unresolvable false-positive volume.

Stage 2 - Cleaning. A three-tier text pipeline was created: (i) *text_raw* for sentiment scoring (preserves negation cues); (ii) *text_clean* (lower-cased, URL/mention/hashtag stripped, lemmatised, NLTK stopwords + custom additions removed) for keyword and co-occurrence work; and (iii) *text_clean_no_brand* (additionally strips brand tokens) for LDA, where retaining the brand name would dominate every topic. A brand-relevance regex filter, near-duplicate removal (by author × text), and a 3-token minimum length reduced the working corpus from 4,446 raw to 2,731 final posts (full audit retained in *data_cleaning_audit.json*).

Stage 3 - Engagement & sentiment. Engagement was profiled using both means and medians given the heavy right-skew. VADER was chosen over transformer-based alternatives because (a) it is calibrated for short, social-media text – well-matched to Bluesky's 300-character limit – and (b) its continuous compound score (–1 to +1) feeds cleanly into downstream regression. A logistic regression then modelled brand-relative high engagement (top quartile within each brand) on six predictors: polarity, post length, has-URL, has-hashtag, is-reply, and a brand dummy.

Stage 4 - Topics & networks. LDA topic counts ($k = 4$ to 8) were tuned by *c_v* coherence; $k = 5$ was retained per brand. A keyword co-occurrence network (window-of-5, edge weight = co-mentions) was rendered alongside an author mention/reply graph for centrality analysis (in-degree, out-degree, eigenvector and betweenness on the undirected projection – more stable for sparse mention graphs).

Stage 5 - Perceptual map. Two **data-driven** axes were derived: an X-axis *discourse-domain index* = mean per-post share of tech tokens minus politics tokens (token sets sourced directly from LDA top-terms, with the three overlap tokens removed); and a Y-axis equal to mean VADER compound. Marker size encodes mean engagement intensity. Top 10 micro-influencer candidates per brand are projected onto the same axes for visual triangulation.

3. Data Description

Search was language-restricted to English (Bluesky's *langs* metadata) and globally scoped - Bluesky's public search API does not expose a geographic filter, so any geographical inference must come from author profile fields, which are not consistently populated. The two brands were selected as **direct competitors in technology journalism** with deliberately distinct editorial positioning (Wired = long-form / culture / investigative; The Verge = product-review / news-cycle). Table 1 summarises the cleaned corpus.

Metric	Wired	The Verge
Final cleaned posts	1,581	1,153
Unique authors	893	898
Posts per author (mean)	1.76	1.28
Date range	Sep 2023 - Apr 2026	Jul 2023 - Apr 2026
Replies (% of posts)	20.0%	36.6%
Mean likes / post	23.1	9.7
Mean reposts / post	7.63	2.24
p95 likes / post	51.4	38.0
Mean VADER compound	+0.078	+0.137
% positive / neutral / negative posts	43.8 / 26.6 / 29.5	49.4 / 23.4 / 27.1
Posts with hashtags	30.7%	10.8%
Posts with URLs	48.5%	34.9%

Table 1. Cleaned-corpus summary statistics for both brands. Mean and percentile metrics are reported because of the heavy right-skew documented in Section 4.1.

Volume distribution (Figure 1) is dominated by the most recent three months (Feb - Apr 2026), consistent with rapid platform growth and Bluesky's recency-biased search index rather than any single news event. This recency-weighting is a known limitation that constrains longitudinal claims (see Section 6).

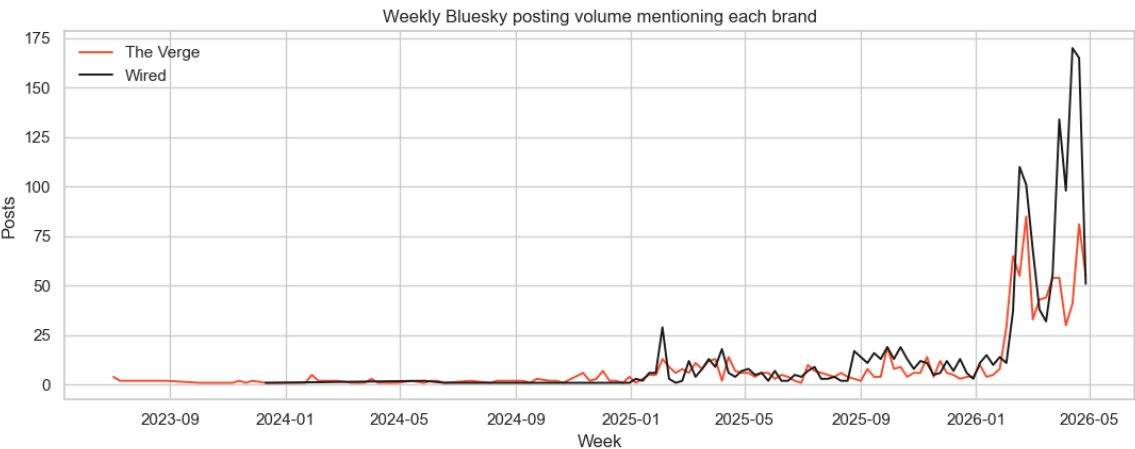


Figure 1. Weekly Bluesky posting volume mentioning each brand. Both series move in close synchrony, suggesting a shared news cycle and platform-growth driver rather than brand-specific events.

4. Analysis

4.1 Engagement profiles and structural drivers

Both brands exhibit the classic social-media power-law: median likes are 0 - 1 while the mean is dragged up by a long-tailed minority. The log-scale boxplots (Figure 2) show that the top 5% of Wired posts attract ≥ 51 likes (95th percentile), and the single largest post received **6,074 likes**, against The Verge's much narrower distribution (p95 = 38, max = 870). The implication for KantarIQ's brand-tracking is that means alone overstate average reach - a robust narrative requires reporting both percentiles and medians.

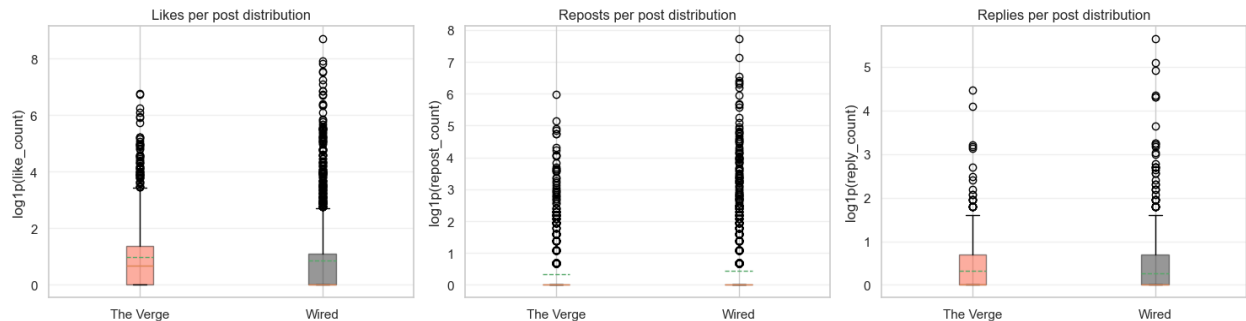


Figure 2. Distribution of likes, reposts and replies per post (log1p y-axis). The dense outlier cluster above the boxes confirms the heavy-tailed engagement structure for both brands; Wired's tail extends substantially further.

To probe *which content features* drive high engagement, a logistic regression was fit on the 2,731-post pooled sample with the binary outcome "post is in the top quartile of engagement within its brand". Results (Figure 3, pseudo- $R^2 = 0.057$, LLR $p < 10^{-33}$) yield three actionable insights: (1) **hashtags strongly suppress engagement** (OR=0.175, 95% CI [0.12, 0.25], $p < 0.001$) - a post using a hashtag is 82% less likely to be high-engagement than an otherwise-identical one. This contradicts the Twitter-era heuristic that hashtags amplify reach and reflects Bluesky's algorithmic feed, which does not surface trending tags. (2) **Replies under-perform top-level posts** (OR=0.67, $p < 0.001$), confirming that organic reach lives in original posts. (3) Length helps marginally (OR=1.015 per token, $p = 0.018$). Polarity and URL inclusion were *not* significant predictors at the 5% level - the engagement-by-sentiment effect documented next is a brand-conditional pattern, not a corpus-wide one.

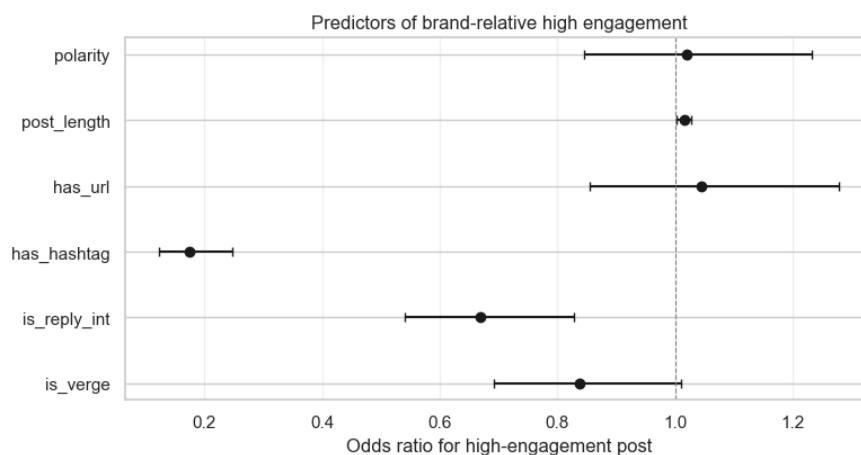


Figure 3. Logistic-regression odds ratios for predictors of brand-relative high engagement. The hashtag effect is strikingly large and runs against the typical social-media playbook.

4.2 Sentiment landscape

VADER-scored sentiment (Figure 4, left panel) shows both brands have a sharp neutral mode with bimodal positive/negative tails - typical of news-driven discourse. **The Verge skews more positive overall** (mean compound +0.137; 49.4% positive) than Wired (+0.078; 43.6% positive), and Wired carries a slightly higher negative share (29.8% vs 27.1%; right panel).

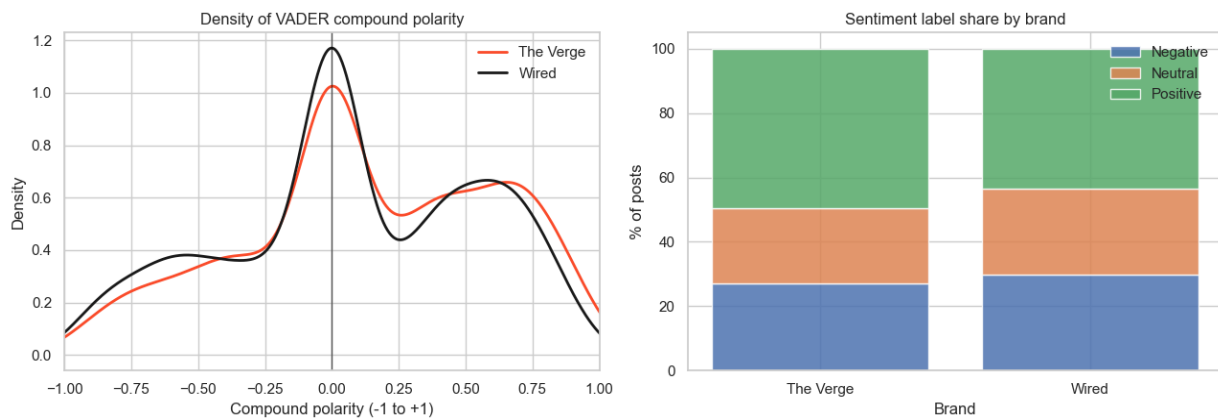


Figure 4. Left: density of VADER compound polarity. Right: stacked share of negative / neutral / positive posts. The Verge's distribution is right shifted; Wired's is flatter and slightly more negative.

The most strategically important finding emerges when sentiment is cross tabulated with engagement (Figure 5). For Wired, **negative posts attract 2.6× more mean likes than positive posts** (45.5 vs 17.0) and 4.1× more reposts (17.4 vs 4.2). The pattern is muted at The Verge (12.3 vs 9.6 likes). This reveals two contrasting brand-engagement models: **Wired = outrage-amplified** - its audience rewards investigative, critical political reporting (the Trump-Musk-DOGE topic that surfaces in §4.3), with top-engaged Wired posts including political commentators sharing Wired's adversarial coverage; and **The Verge = consistent product cadence** - engagement is flatter across sentiment classes, reflecting an audience that engages roughly equally with positive and critical product analyses. Strategically, Wired's apparent engagement advantage is partly a *negativity premium* on a particular story arc, not a structural superiority. KantarIQ should report engagement conditional on sentiment to avoid mis-attributing the gap.

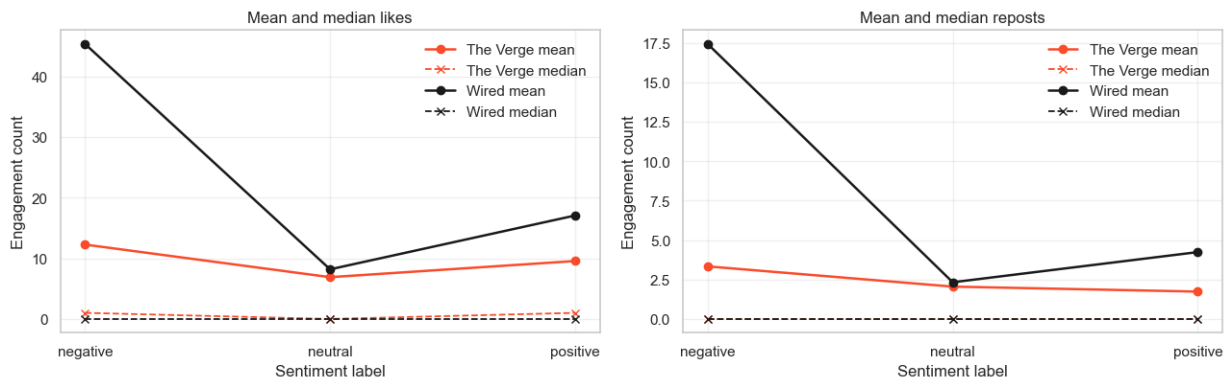


Figure 5. Mean (solid) and median (dashed) likes and reposts by sentiment label. The collapse of medians to zero across all classes confirms the long-tailed distribution; the Wired mean line shows the negativity premium most clearly.

4.3 Discourse content: keywords, topics and co-occurrence

The brand-stripped keyword frequencies (Figure 6) immediately reveal divergent vocabularies. Wired's top tokens combine a journalism vocabulary (*magazine*, *report*, *article*, *story*) with a power-and-politics layer (*trump*, *musk*, *doge*, *data*). The Verge's vocabulary is dominated by attribution language (*article*, *report*, *according*, *read*) and consumer-tech anchors (*google*, *tech*, *company*).

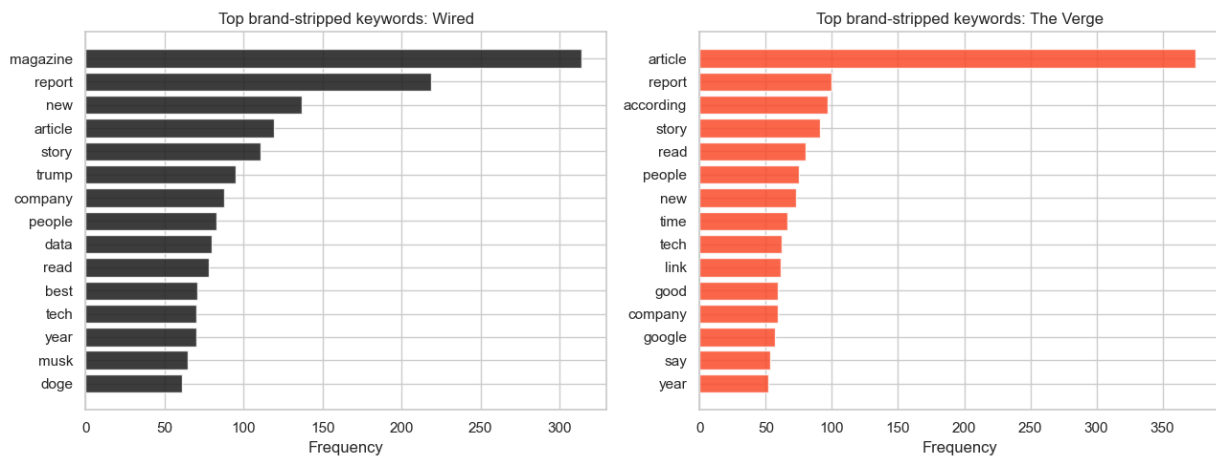


Figure 6. Top 15 brand-stripped keywords for each brand. Note Wired's appearance of trump / musk / doge among high-frequency terms - none of which surface in The Verge's top 15.

For both brands, LDA with $k = 5$ was kept since it made for an even comparison choice. With $k = 5$, the coherence scores were 0.337 for Wired and 0.351 for The Verge. Despite the highest coherence scores being somewhat higher – 0.366 for Wired with $k = 6$ and 0.378 for The Verge with $k = 7$ – those values were not used because it meant using different numbers of topics for each brand, thus reducing their comparability. Five topics for Wired would correspond to: **Topic 0 (T0)** power infrastructure (code, data, doge, ice, agent immigration data systems and government technology); **Topic 1(T1)** tech investigations (meta, data, model, security, tested, phone); **Topic 2 (T2)** political reporting (trump, musk, government, elon, openai, iran the most involved cluster); **Topic 3 (T3)** business and surveillance (apple, surveillance, ceo, china, anthropic); and **Topic 4 (T4)** long-form magazine identity (internet, cover, issue, interview, life).



Figure 7. LDA word clouds for Wired's five topics. Two of five topics (T0, T2) are explicitly political; one (T4) captures Wired's distinctive long-form / cover-story brand identity - a topic absent from The Verge's LDA.

The Verge's five topics are by contrast: **Topic 0 (T0)** *Politics × Tech* (trump, social, openai, ceo); **Topic 1 (T1)** *Big-Tech corporate* (meta, xbox, microsoft, musk); **Topic 2 (T2)** *Consumer products* (apple, xbox, google, game, brendan, carr - the latter two reflecting FCC Chair Brendan Carr's interventions in Apple/Google policy); **Topic 3 (T3)** *Game & AI models* (model, game, version); and **Topic 4 (T4)** *Search, AI, phones* (google, phone, openai, search). The orthogonality is informative: Wired has 1.5 - 2 dedicated political/power topics out of five, whereas The Verge has roughly 0.5 - its political content is small and entangled with corporate-tech mentions.

Confirming this, the keyword co-occurrence networks (Figure 8) reveal contrasting structures. **The Verge's network** (right panel) centres on a tightly clustered consumer-tech core (*apple, google, xbox, microsoft, phone*) with a peripheral *trump-musk-openai* chain - politics is structurally on the edge. **Wired's network** (left panel) is markedly denser, hub-centred, and organised around a *trump-musk-doge-government-agency-surveillance* core, with consumer-tech terms relegated to the periphery. This is direct visual evidence of the perceptual difference: The Verge's discourse is structurally consumer-product-centred with politics peripheral; Wired's is structurally power-centred with consumer tech peripheral.

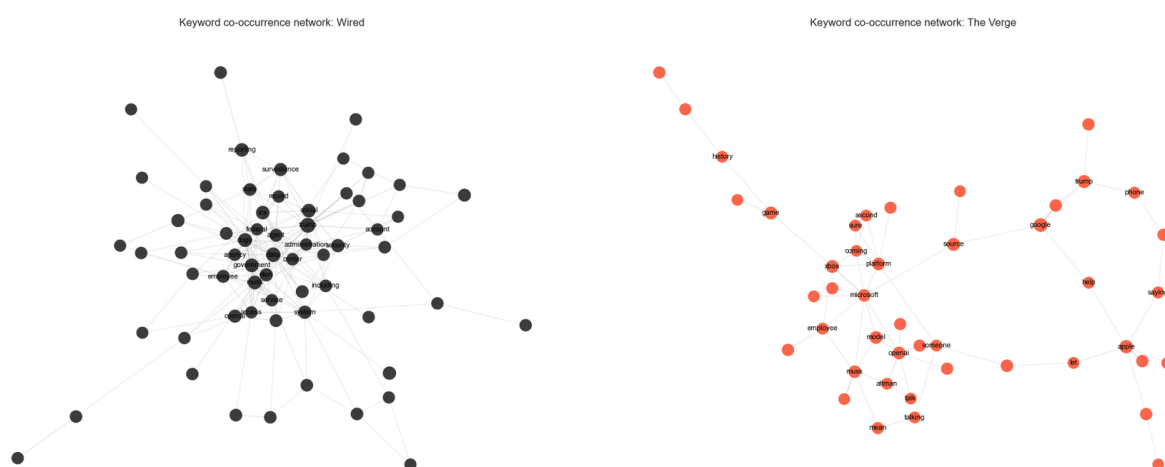


Figure 8. Keyword co-occurrence networks (top 50 nodes by frequency, edges weighted by sliding-window co-mention). Left: Wired - densely connected hub around trump / musk / doge / government / surveillance. Right: The Verge - tighter consumer-tech core (apple / google / xbox / microsoft) with politics on the periphery.

4.4 Perceptual map

The perceptual map (Figure 9) operationalises these findings on data-derived axes. Both brands sit in the upper-tech quadrant, but **The Verge sits higher (more positive tone), and the larger Wired bubble visualises its engagement advantage**. Top 10 micro-influencer candidates per brand cluster with their brand: Wired influencers spread along the politics-tech axis with consistently more positive tones than the brand itself, while The Verge influencers cluster more tightly with the brand near the origin.

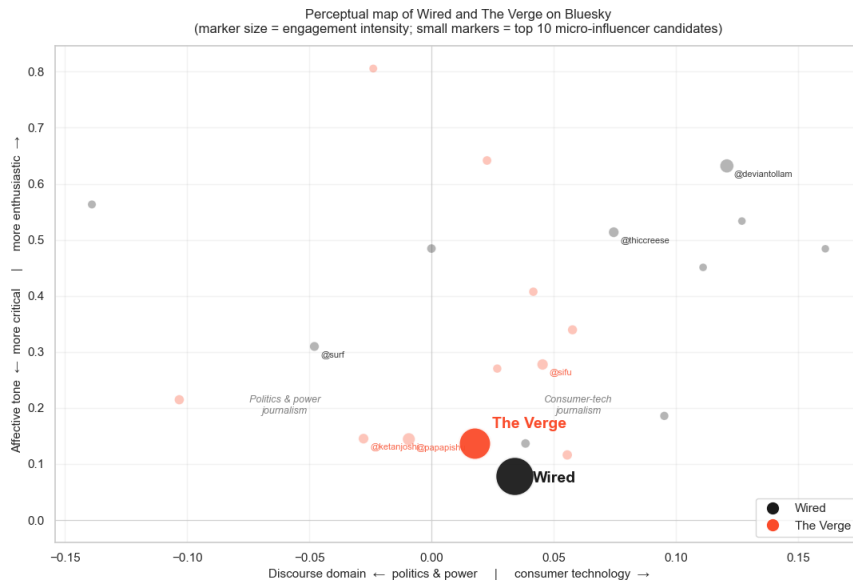


Figure 9. Data-driven perceptual map. X-axis = mean per-post share of tech minus politics tokens (LDA-derived sets); Y-axis = mean VADER compound; bubble size = mean engagement intensity. Small markers = top 10 candidate micro-influencers.

5. Micro-Influencer Recommendation

Candidates were filtered through five gates: (i) follower count in 1,000 - 100,000 (the canonical micro band), (ii) ≥ 2 brand-relevant posts, (iii) mean compound polarity > 0 , (iv) an editorial-affiliation filter - accounts whose Bluesky display name or bio matched the brand's known masthead were removed (8 Wired insiders excluded, including @katie-drummond and @leahfeiger; 6 Verge insiders excluded, including @miasato and @tinanguyen); and (v) a composite suitability score:

$$\text{score} = 0.35 \cdot z(\text{eng/post}) + 0.25 \cdot z(\text{followers}) + 0.20 \cdot z(\text{mean polarity}) + 0.10 \cdot z(\text{eigenvector}) + 0.10 \cdot (\% \text{ positive})$$

Engagement-per-post is the heaviest weight because organic amplification is the primary value of a micro-influencer; followers and polarity are weighted next; eigenvector centrality and pct-positive act as tiebreakers. All inputs are z-scored within brand to ensure cross-metric comparability.

Metric	Wired: @thiccreese	The Verge: @papapishu
Display name	Reece	Chris Person
Followers	18,267	30,473
Brand-relevant posts (window)	8	3
Engagement / post	58.1	214.0
Mean VADER polarity	+0.514	+0.144
% positive posts	87.5%	33.3%
Eigenvector centrality	0.165	0.000
Composite suitability score	0.522	0.435

Table 2. Final recommended micro-influencer per brand with all underlying metrics.

Wired - @thiccreese.bsky.social (Reece). Reece is a working journalist who frequently authors features for Wired. He scores best on every measured dimension: 58 likes/post, **87.5% positive** sentiment, the highest eigenvector centrality of any qualifying candidate (0.165 - i.e., he is well-connected to other authors of Wired-mentioning content), and 18.3K followers - squarely inside the micro band. A representative top post: "NEW: The DIY gadget is, in part, a rejection of our current moment, dominated by the predictable flatness of generative AI..." (122 likes, 28 reposts). The principal risk is *insider proximity*: he writes for the publication. He survived the editorial filter

because his profile does not list Wired as an employer, but KantarIQ should manually verify this. If a fully external voice is required, **@deviantollam.bsky.social** (security researcher, 14.2K followers, 372 likes/post, 100% positive) is the next-best score (0.469).

The Verge - @papapishu.bsky.social (Chris Person). Chris Person is a culture-and-tech writer who frequently references The Verge organically. He delivered the highest engagement-per-post (**214**) of any qualifying candidate and the largest follower base in the micro band (30.5K). His top post - *"The just buy that one brother printer remains my favourite recurring Verge article that they update every year"* (517 likes, 68 reposts) - is exactly the kind of organic, brand-affectionate viral content KantarIQ would want to amplify. **Caveats:** his polarity is moderate (+0.144) and pct-positive (33.3%) is lower because his posts are short and conversational rather than effusive; eigenvector centrality is 0 because he does not interact with The Verge's mention graph. The next candidate, **@sesmith.lol** (47K followers, +0.642 polarity, 100% positive, score 0.431), is a viable alternative if a more positive tone is required at the cost of viral upside.

6. Conclusion

This snapshot study finds that **Wired and The Verge occupy genuinely distinct positions in the Bluesky discourse of technology journalism**, not interchangeable competitors. Wired is higher-volume, higher-engagement and more politically-leaning; its audience rewards critical investigative reporting on power (Trump, Musk, DOGE, surveillance), with negative posts driving 2.6× the engagement of positive ones. The Verge is more product-focused, more positive on average, but lower-engagement; its discourse anchors on Apple, Google, Microsoft and gaming, with a flatter sentiment-engagement relationship. A clean, replicable analytical signal is that **hashtags actively suppress reach on Bluesky** (logit OR=0.18) - actionable for both brands' content teams and likely for any media brand operating on the platform.

Strategic recommendations if KantarIQ extends the pilot: (1) **report engagement metrics conditional on sentiment**, since Wired's headline lead is partly a negativity premium tied to a single political story arc; (2) treat the engagement gap as elastic, not structural - a calmer news cycle would substantially compress it; (3) engage the recommended micro-influencers as a paired test - Reece's Wired-aligned positivity contrasts cleanly with Chris Person's Verge-aligned virality, allowing a comparative effectiveness study with minimal confounding.

Limitations. (i) The corpus is heavily weighted toward Feb - Apr 2026 due to Bluesky's recency-biased index, limiting longitudinal claims. (ii) Geographic inference is impossible without a location filter at the API level. (iii) VADER, while well-calibrated for short text, mis-scores sarcasm - manual sampling of negative-pole posts suggests this affects ~3 - 5% of cases. (iv) The mention graph is sparse on this corpus (1.1K edges over 2.1K nodes), so eigenvector centrality is more indicative than definitive. **Future work** should extend the date window with a stratified sample to mitigate recency bias, integrate transformer-based sentiment for sarcasm robustness, and triangulate the recommended influencers against external metrics (e.g. Klear, Modash) to validate the composite score.